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Brocade 64Gb Fibre Channel Module for HPE Synergy

QuickSpecs

High-Performance 64Gb SAN Module for Synergy.

The Brocade 64Gb Fibre Channel Module for HPE Synergy represents a composable and integrated Fibre Channel interconnect module with Gen7 technology that simplifies integration of the HPE Synergy blade chassis into a Storage Area Network (SAN). It provides breakthrough application performance, operational stability, and simplified manageability to meet the stringent business requirements with 64Gb FC technology for HPE's Composable Infrastructure. It is ideal for enterprises planning to deploy 64Gb FC to meet the ever-growing bandwidth demands of high-density virtualization/applications utilizing top-end Storage arrays and for customers looking to deploy scalable private cloud storage networks using HPE Synergy for workloads.

Brocade 64Gb Fibre Channel Module for HPE Synergy

The Brocade 64Gb FC Switch for HPE Synergy provides industry-leading performance with <700ns frame latency for fast switching, switching bandwidth up to 1,536 Gbps to enable demanding applications and high-density server virtualization. It provides scalability with 12 external SFP+ ports, each supporting 64Gb, 32Gb, and 16Gb FC speeds that customers can utilize on-demand paralleling their business growth. The Brocade 64Gb FC Module for HPE Synergy includes HPE Power Pack+ SAN management software baseline, which enables customers to manage data flow and maintain optimized application performance. With the HPE PowerPack+ software, customers can also utilize IO Insight software for proactive and real-time monitoring of storage and application-level IO and VM Insight software to optimize VM workloads and ensure service-level objectives are met. Brocade 64Gb FC Switch Module for HPE Synergy is designed with NVMe in mind and provides next-generation storage connectivity to existing storage networks. Brocade 64Gb FC Module for HPE Synergy also lowers TCO as it hot-plugs into the back of the HPE Synergy Frame. This frees-up rack space, enables shared power and cooling, and reduces cabling, resulting in significant infrastructure savings.



Brocade 64Gb Fibre Channel Module for HPE Synergy

Models

Description	SKU
Brocade 64Gb Fibre Channel Module for HPE Synergy	P77653-B21

At a Glance

Integrated SAN connectivity

Enables data center consolidation with HPE Synergy through a state-of-the-art zero-footprint design while lowering cost through shared power and cooling, fewer cables and less space resulting in reduced infrastructure complexity and lower operations costs.

64 Gb FC storage networking

Enables support of virtualized environments and other emerging technologies through high-speed technology that doubles the performance of the storage network. External facing ports auto-sense lower bandwidth components for flexibility and investment protection with existing FC ports (64Gb, 32Gb, and 16Gb FC for external ports) as well as to the future FC products.

Non-blocking architecture

Includes connection speed support for all ports to provide a high-performance system with switching bandwidth of up to 1,536 Gbps (24 ports x 64 Gbps).

ISL Trunking performance enhancements

Support for multiple trunks, with the maximum trunk size of up to eight SAN ports for up to 512Gbps balanced throughput per each trunk group. Trunking of the SAN-facing ports with external Brocade or HPE B-series switches provides a high-bandwidth solution while lowering costs through fewer SFPs and cables. You can trunk up to 8 external FC SFP+ ports.

Fabric Vision

Enables powerful built-in monitoring, SAN management, and diagnostic tools to simplify administration, increase uptime and reduce costs. This results in significant insight and visibility across SANs with extensive analytics to keep your SAN optimized for best results. Available by default in the Power Pack+ model or as an upgrade option. It enables IO Insight and VM Insight Software.

Access Gateway mode

Employs N_Port ID Virtualization (NPIV), an ANSI T11 standard, to enable interoperability with other SAN fabrics and eliminate domain considerations while improving SAN scalability.

Broad range of management tools and SMI-s conformance

Integrates with HPE OneView for monitoring and leverages HPE B-series SAN management and monitoring tools (PowerPack+, Fabric Vision). New REST API in addition to existing SMI-s API enables the use of industry standard interfaces to integrate management with other applications.

Utilizes advanced Web Tools to simplify fabric set-up and on-going maintenance for SAN administrators.

What's New

- New 64 Gb/s Fibre Channel embedded high-performance switch module for HPE Synergy Frame
- 12 downlink internal server ports, up to 12 uplink (SFP+) external ports

Advanced Fabric Services

- Hardware Enforced Zoning*
- Dynamic Path Selection*
- Web Tools
- Power Pack+ fabric services software bundle (included)
  - Extended Fabrics*
  - Fabric Vision (Includes IO Insight and VM Insight Software)
  - ISL Trunking

Notes: *Not supported in Brocade Access Gateway mode.

Manageability

- Web Tools
- HPE Storage Operations Orchestration (optional)
- SMI API
- SNMP
- REST API
- SAN automation with Ansible

Interoperability

Brocade Access Gateway mode enables the Brocade 64Gb Fibre Channel Module for HPE Synergy to interoperate with other SAN fabrics running supported firmware. While in Access Gateway mode, the device must be connected to an NPIV-enabled edge switch or director. Supported edge environments are listed in the SAN Design Guide https://support.hpe.com/hpsc/doc/public/display?docId=c00403562

Software Components, Standard

Access Gateway*

Brocade Access Gateway Mode enables seamless connectivity for the Brocade 64Gb Fibre Channel Module for HPE Synergy to other supported SAN fabrics, enhances scalability and simplifies manageability. Utilizing N_Port ID Virtualization (NPIV) standards, Brocade Access Gateway eliminates switch/domain considerations.

Notes: *Brocade Access Gateway mode does not support switch-related features, including some features in the Power Pack+ bundle (e.g., Extended Fabric). Please refer to the Access Gateway Administrator's Guide for details.

Frame Filtering

A special B-Series ASIC-based capability that enables new applications and features. The switch has the ability to "view" the first 64-bytes of the Fibre Channel frame. At this time, Frame Filtering enables advanced capabilities such as Advanced Zoning, Frame-level analytics and statistics.

Advanced Zoning

Zoning (Port and Worldwide Name), as well as Access Control, are enforced by hardware within the ASIC. Hardware enforcement enables the highest level of zoning enforcement without any impact on performance. Administrators can organize a physical fabric into logical groups and prevent unauthorized access by devices outside the Zone.

Web Tools
Enables organizations to monitor and manage single Fibre Channel switches and monitor small SAN fabrics. Tasks can be performed through a Web browser from a standard laptop, desktop PC, or workstation from any location within the enterprise.

Dynamic Path Selection
Improves performance by dynamically routing data traffic across multiple links and trunk groups.

Adaptive Networking Services
Adaptive Networking services optimize fabric behavior and ensure ample bandwidth for mission-critical applications. The tools currently include Quality of Service (QoS), Ingress Rate Limiting and Traffic Isolation. By assigning zones a high, medium, or low priority, QoS allocates bandwidth in the event of congestion to expedite high-priority traffic and keep all traffic flowing. Ingress Rate Limiting restricts data flow from less-critical hosts at preset bandwidths. Traffic Isolation assigns high-bandwidth data flows to specific ISLs.

Notes: Adaptive Networking Services are included by default in the switch firmware and do not require a separate license.

Software Components, Optional

Power Pack+ Software Bundle
The Power Pack+ Software Bundle includes:

- Fabric Vision (Includes IO Insight and VM Insight Software)
- ISL Trunking
- Extended Fabrics*

Notes:

- *Not supported in Access Gateway mode (see the Access Gateway Administrator´s guide for full details).
- Adaptive Networking and Server Application Optimization features are included by default in the switch firmware and do not require a separate license.

Fabric Vision
Fabric Vision offers innovative diagnostic, monitoring and management capabilities to help accelerate new application deployments, address SAN problems before they impact operations and reduce operational costs, it includes:

- VM Insight: Seamlessly monitors end-to-end VM performance through standards-based VM tagging.
- IO Insight: Automatically detects degraded storage IO performance with integrated device latency and IOPS monitoring.
- Monitoring and Alerting Policy Suite: A policy-based monitoring tool with pre-built rules and automation that simplifies fabric-wide threshold configuration and monitoring. In addition, administrators can include IO Insight metrics in MAPS policies to understand IO profile as well as to be notified of storage IO performance degradation.
- Configuration and Operational Monitoring Policy Automation Services Suite (COMPASS): Simplifies deployment, safeguards consistency and increases operational efficiencies of larger environments with automated switch and fabric configuration services. Administrators can configure a template or adopt an existing configuration to seamlessly deploy a configuration across the fabric.
- ClearLink® Diagnostics: Ensures optical and signal integrity for Fibre Channel optics and cables, simplifying deployment and support of high-performance fabrics. ClearLink Diagnostic.

- Port (D_Port) is an advanced capability of Fibre Channel platforms.
- Flow Vision: A comprehensive tool that enables administrators to identify, monitor, and analyze specific application data flows without using taps to ensure optimized performance.
- Forward Error Correction (FEC): Enables recovery from bit errors in device connections and ISLs, enhancing transmission reliability and performance.

HPE SANnav Management Software

HPE SANnav Management Software is the next-generation SAN management application suite for HPE B-series SAN environments. It consists of SANnav Management Portal Software and SANnav Global View Software:

- SANnav Management Portal is a next-generation SAN management application with a simple browser-based user interface (UI) streamlining common workflows, such as configuration, zoning, deployment, troubleshooting, and reporting.
- SANnav Global View helps administrators visualize the health, performance and inventory of multiple SANnav Management Portal instances at data centers across the globe or a single multi-tenant data center using a simple, intelligent dashboard.

SANnav Management Portal and SANnav Global View not only transform SAN telemetry data into useful insights, such as health and performance scores, but also enable administrators to quickly associate real-time data with historical metrics and logs for in-depth analysis. This can help with spotting trends, establishing baselines, and identifying any behavioral changes over time.

HPE SANnav Management Software is available as a term-license for 1-year, 3-years and 5-years period as both – physical and electronic License-to-Use (LTU). It supports 16Gb, 32Gb, and 64Gb FC Switches and Directors.

ISL Trunking

For high performance and seamless failover, this optional license logically groups up to eight E_ports (switch mode) or F_ports (Access Gateway mode) to provide a high bandwidth trunk between two HPE B-Series or Brocade switches. Each switch needs its own license. The fabric operating system views the trunk as a single, high bandwidth ISL resource (up to 512 Gbps when routing connections between 64 Gb/s capable switches). Connections are load-balanced across the individual links, which comprise the logical trunk group.

HPE Synergy Frame 12000 compatibility

Brocade 64Gb FC Module for HPE Synergy must be deployed within the HPE's Composable Infrastructure solution and is compatible only with HPE Synergy Frames.

HPE FC HBA Mezzanines

64Gb and 32Gb HPE Fibre Channel Mezzanine cards for HPE Synergy are supported

- Please refer to the HPE FC Mezzanine HBA Product QuickSpecs for Compute Module support details.

HPE Synergy Frame

HPE Synergy Frame 12000

Switch interoperability

HPE B-Series Switches, Directors, and Routers

64Gbps Fibre Channel Switches and Directors

- HPE SN8700B 64Gb Director Switch (Enterprise)
- HPE SN6750B 64Gb Fibre Channel Switch
- HPE SN6700B 64Gb Fibre Channel Switch (Midmarket)
- HPE SN3700B 64Gb Fibre Channel Switch (Entry)
- HPE Synergy 7330C 64Gb Fibre Channel Host Bus Adapter
- Brocade 64Gb Fibre Channel Module for HPE Synergy

32Gbps Fibre Channel Switches and Directors

- HPE SN8600B 32Gb Director Switch (Enterprise)
- HPE SN6650B 32Gb Fibre Channel SAN Switch
- HPE SN6600B 32Gb Fibre Channel Switch (Midmarket)
- HPE SN3600B 32Gb Fibre Channel Switch (Entry)
- HPE Synergy 5331C 32Gb Fibre Channel Host Bus Adapter
- HPE Synergy 5330C 32Gb Fibre Channel Host Bus Adapter
- HPE Synergy 5830C 32Gb Fibre Channel Host Bus Adapter
- Brocade 32Gb Fibre Channel SAN Switch Module for HPE Synergy

16Gbps Fibre Channel Switches and Directors

- HPE SN8000B 16Gb Director Switch (Enterprise)
- HPE SN6500B 16Gb Fibre Channel SAN Switch
- HPE SN6000B 16Gb Fibre Channel Switch (Midmarket)
- HPE SN3000B 16Gb Fibre Channel Switch (Entry)
- Brocade 16Gb Fibre Channel SAN Switch Module for HPE Synergy

Standard Features

Virtual Connect

- HPE Virtual Connect 32 Gb 24-port Fibre Channel Module
- HPE Virtual Connect 16 Gb 24-port Fibre Channel Module

Multi-Protocol Routers and Gateways

- HPE SN4700B 64Gb Fibre Channel SAN Extension Switch
- HPE B-series SN8600B 32Gb SAN Extension Blade
- HPE Multi-protocol Extension Blade
- HPE B-series SN4000B FC SAN Extension Switch
- HPE 1606 SAN Extension Switch

Standard Features

Family Comparison

Features	Brocade 16Gb Fibre Channel SAN Switch Module for HPE Synergy	Brocade 32Gb Fibre Channel SAN Switch Module for HPE Synergy	Brocade 64Gb Fibre Channel Module for HPE Synergy
Targeted Environment	Enterprise, Datacenters, Workgroups, Departments	Enterprise, Datacenters, Workgroups, Departments	Enterprise, Datacenters, Workgroups, Departments
Port Bandwidth	16Gbps	32Gbps	64Gbps
Aggregate device bandwidth	576 Gbps (36 ports x 16 Gbps)	896 Gbps (28 ports x 32 Gbps)	1,536 Gbps (24 ports x 64 Gbps)
OS Support	See SPOCK https://www.hpe.com/storage/spock		
Storage system Support	HPE 3Par StoreServ P8400, MSA 2052/MSA2060, Nimble Series HF, XP7/XP8, Primera C630, Alletra 6000/6030/9000/9060		HPE 3Par StoreServ P8400, MSA 2052/MSA2060, Nimble Series HF, XP7/XP8, Primera C630, Alletra 6000/6030/9000/9060
FC Ports	8 SFP+ external, 4 QSFP external /12 internal (24 maximum)	8 SFP+ external, 2 QSFP external /12 internal	12 SFP+ external /12 internal
SFP	B-series	B-series	B-series
Advanced Trunking	Included with Power Pack+ or Optional Upgrade	Included with Power Pack+ or Optional Upgrade	Included with Power Pack+
Adaptive Networking	Included	Included	Included
Fabric Vision Software	Included with HPE PowerPack+ upgrade	Included with HPE PowerPack+ upgrade	Included with HPE PowerPack+
Form factor	Embedded	Embedded	Embedded
Zoning Software	Yes (Included)	Yes (Included)	Yes (Included)
Hot plug, redundant power supplies	Yes, in Synergy Frame	Yes, in Synergy Frame	Yes, in Synergy Frame
Hot plug fans	Yes, in Synergy Frame	Yes, in Synergy Frame	Yes, in Synergy Frame

Service and Support

For the most up-to-date information on HPE Services, please refer to the HPE Services – Supplemental QuickSpecs, which provides a comprehensive and regularly updated overview of available services.

Configuration Information

Step 1 – Base Configuration

Description	SKU
Brocade 64Gb Fibre Channel Module for HPE Synergy	P77653-B21

- (1) 64Gbps SAN Switch
- 24 ports enabled for any combination (12 internal and 12 external)
- 4 64Gb SFP56 Transceivers (R7M15A)
- Dual function mode - Native OS (default) or Access Gateway Mode
- Documentation and includes Power Pack+ Bundle and management tools (see software components)

Step 2 – Additional Options

Select each type of required options with quantities specified.

Hardware

64Gbps / 32Gbps Optical Transceivers (SFP+)

Description	SKU
HPE B-series 64Gb SFP56 Short Wave 1-pack Secure Transceiver	R7M15A
HPE B-series 64Gb SFP56 Short Wave 8-pack Secure Transceiver	R7M16A
HPE B-series 32Gb SFP28 Short Wave 1-pack Secure Transceiver	R6B12A
HPE B-series 32Gb SFP28 Short Wave 8-pack Secure Transceiver	R6W26A
HPE B-series 16Gb SFP+ Short Wave Transceiver	QK724A
HPE B-series 16Gb SFP+ Short Wave 8-pack Secure Transceiver	R6W28A

Optical cables

PremierFlex OM4 LC/LC Multi-mode Cables

HPE Premier Flex LC/LC Multi-mode OM4 2 Fiber 1m Cable	QK732A
HPE Premier Flex LC/LC Multi-mode OM4 2 Fiber 2m Cable	QK733A
HPE Premier Flex LC/LC Multi-mode OM4 2 Fiber 5m Cable	QK734A
HPE Premier Flex LC/LC Multi-mode OM4 2 Fiber 15m Cable	QK735A
HPE Premier Flex LC/LC Multi-mode OM4 2 Fiber 30m Cable	QK736A
HPE Premier Flex LC/LC Multi-mode OM4 2 Fiber 50m Cable	QK737A

OM3 LC/LC Multi-mode Cables

HPE LC to LC Multi-mode OM3 2-Fiber 0.5m 1-Pack Fiber Optic Cable	AJ833A
HPE LC to LC Multi-mode OM3 2-Fiber 1.0m 1-Pack Fiber Optic Cable	AJ834A
HPE LC to LC Multi-mode OM3 2-Fiber 2.0m 1-Pack Fiber Optic Cable	AJ835A
HPE LC to LC Multi-mode OM3 2-Fiber 5.0m 1-Pack Fiber Optic Cable	AJ836A
HPE LC to LC Multi-mode OM3 2-Fiber 15.0m 1-Pack Fiber Optic Cable	AJ837A
HPE LC to LC Multi-mode OM3 2-Fiber 30.0m 1-Pack Fiber Optic Cable	AJ838A
HPE LC to LC Multi-mode OM3 2-Fiber 50.0m 1-Pack Fiber Optic Cable	AJ839A

Software

Power Pack+ Software Included

Includes bundled license for:

- Fabric Vision (Includes IO Insight and VM Insight Software)
- ISL Trunking
- Extended Fabrics

SANnav Software Licenses:

Notes: For users purchasing SANnav licenses for the first time, one of the following SKUs should be purchased. For users wanted to renew a SANnav license, a renewal license should be selected from the SANnav Renewal Software License in the next section.

HPE SANnav Management Portal Base 1yr E-LTU	R3P45AAE
HPE SANnav Management Portal Enterprise 1yr E-LTU	R3P46AAE
HPE SANnav Global View 1yr E-LTU	R3P47AAE
HPE SANnav Management Portal Base 3yr E-LTU	R3P48AAE
HPE SANnav Management Portal Enterprise 3yr E-LTU	R3P49AAE
Description	**SKU**
HPE SANnav Global View 3yr E-LTU	R3P50AAE
HPE SANnav Management Portal Base 5yr E-LTU	R4P29AAE
HPE SANnav Management Portal Enterprise 5yr E-LTU	R4P30AAE
HPE SANnav Global View 5yr E-LTU	R4P31AAE

Notes:

- The Base edition of HPE SANnav Management Portal does not support management of director class switches.
- HPE SANnav Management Software License-to-Use (LTU/E-LTU) includes maintenance and support for the duration of the license. At the end of the license period, customers have the option to renew the license, maintaining the same license key, by selecting from the SANnav Renewal Software License section below Software renewal via HPE Services is not allowed/supported.
- Trial Licenses: depending on the version of SANnav license, the trial period is different.
  - 90-day trial license available with all SANnav releases through v2.2.0.
  - 30-day trial license available with SANnav v2.2.1 and v2.2.2.
  - No trial license available with SANnav v2.3.0 or later.

SANnav Renewal Software Licenses

Notes: For users who currently have a SANnav license and want to renew their software license, one of the following licenses should be selected. The SANnav license must align with the current type of software license i.e., in order to renew a Base license, the Base renewal licenses will need to be selected.

HPE SANnav Management Portal Base 1-year Renewal E-LTU	S1S52AAE
HPE SANnav Management Portal Enterprise 1-year Renewal E-LTU	S1S55AAE
HPE SANnav Global View 1-year Renewal E-LTU	S1S58AAE

Notes:

- The Base edition of HPE SANnav Management Portal does not support management of director class switches.
- HPE SANnav Management Software License-to-Use (LTU/E-LTU) includes maintenance and support for the duration of the license.
- Trial Licenses – depending on the version of SANnav license, the trial period is different.
  - 90-day trial license available with all SANnav releases through v2.2.0.
  - 30-day trial license available with SANnav v2.2.1 and v2.2.2.
  - No trial license available with SANnav v2.3.0 or later.

Fibre Channel ports

12 Internal 32/64Gbps (downlink) ports; 12 External 16/32/64 Gbps auto-sensing (12x SFP+) (uplink) – universal (E, F, N, D) ports.

Device support and Scalability guidelines

https://support.hpe.com/hpsc/doc/public/display?docId=emr_na-c04787660

Performance

64Gbps line speed, full duplex

Certified maximum

Certified maximum information available at
https://support.hpe.com/hpsc/doc/public/display?docId=emr_na-c04787660

Switch core

Non-blocking.

Fabric latency

<=700 nanoseconds through the use of cut-through frame routing at 64Gbps (includes FEC logic delay).

Maximum frame size

2112-byte payload.

Classes of service

Class 2, Class 3, Class F (inter-switch frames).

Port types

D_Port (Diagnostic port), E_Port, F_Port; optional port-type control Brocade Access Gateway mode: F_Port and NPIV-enabled N_Port.

Data traffic types

Fabric switches supporting unicast.

Media types

Hot-pluggable, B-Series Branded Small Form Factor Pluggable (SFP), LC connector; Short-Wave Laser (SWL), Long-Wave Laser (LWL); Extended Long-Wave Laser (ELWL); distance depends on fiber-optic cable and port speed. Supports SFP+ (64/32 Gbps) optical transceivers.

Advanced

Short-wave, long-wave, and extended long wave laser transceivers are supported.

Fabric services

Simple Name Server, Registered State Change Notification (RSCN). Includes Advanced WebTools and Advanced Zoning. Optional advanced fabric services include: Fabric Vision, ISL Trunking, and Extended Fabrics.

Options

SFP media, port upgradeable licensing, advanced management software licensing.

Supported software

Telnet, SSH, SNMP (FE MIB, FC Management MIB), HPE Storage Operations Manager, Advanced WebTools, Advanced Zoning, HPE SANnav Management Software, SmartSAN, and Ansible Automation.

Management access

Web Tools, SMI-s provider, REST API, SNMP, Telnet, and SSH.

Diagnostics

Cable diagnostic (D_Port), including electrical/optical loopback, link traffic/latency/distance.

Power consumption

Typical 50W, maximum is 70W (with all ports running at maximum speed).

Technical Specifications			
Dimensions	**Depth**	1.08 in. (2.75 cm)	
	Height	11.05 in. (28.07 cm)	
	Width	16.92 in. (42.97 cm)	
	Weight	Approx. 7.0 lb. (3.18 kg) - without media	
Environment	**System Inlet Temperature**	**Operating**	Inlet air temperature up to 43°C (109.4°F) at sea level with an altitude derating of 1.0°C per every 1000 feet of elevation above sea level to a maximum operating altitude. For example, the maximum inlet air-temperature requirement at 5000 ft. above sea level is 38°C.
		Non-operating	-40°C to 70°C (-40° to 158°F). Maximum rate of change is 20°C/Hr (36°F/Hr).
	Relative Humidity	**Operating**	5 to 95% relative humidity, 38.7°C maximum wet bulb temperature, and Non-condensing.
		Non-operating	5 to 95% relative humidity (Rh), 38.7°C (101.7°F) maximum wet bulb temperature, non-condensing.
	Altitude	**Operating**	3050 m (10,000 ft.). This value may be limited by the type and number of options installed. Maximum allowable altitude change rate is 457 m/min (1500 ft/min).
		Non-operating	9144 m (30,000 ft.). Maximum allowable altitude change rate is 457 m/min (1500 ft. /min).
Vibration	**Operating**	1 G, 10–500 Hz, 60 minutes per axis	
	Non-operating	2 G, 10–500 Hz, 30 minutes	
Shock	**Operating**	Half-sine-wave shock, 40 G, 2 ms duration, in each direction of three mutually perpendicular axes.	
	Non-operating	Half-sine-wave shock, 140 G, 2 ms duration, in each direction of three mutually perpendicular axes; and square-wave shock up to 80 G, 180 inches/s velocity change, in each direction of three mutually perpendicular axes.	

Summary of Changes

Date	Version History	Action	Description of Change
21-Sep-2026	Version 3	Changed	QuickSpecs maintenance updates.
06-Apr-2026	Version 2	Changed	Added compatibility with 5330C and 5830C 32G HBAs. Added compatible 16G FC switches and 16G SFP+ transceivers. Corrected port counts and quantity of transceivers included.
02-Feb-2026	Version 1	New	New QuickSpecs.

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